What is a learning image in a ViT?

Choose an internal response. Map its signed contributions back to the input.

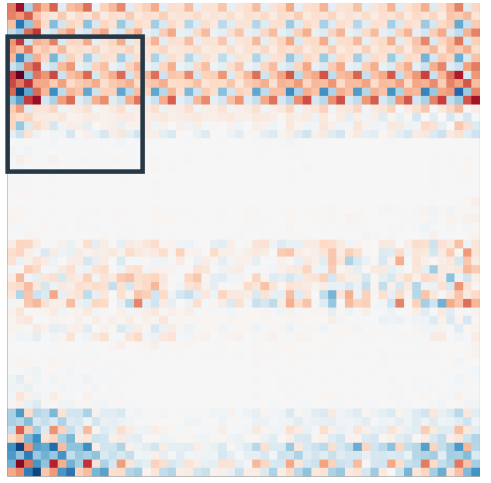


Input image

ViT · Block 0

CLS · Head 10

Selected response

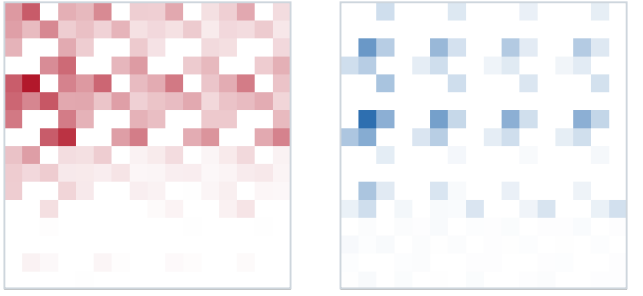


Signed learning image

− +

Shared signed scale · zero at white

Same region, enlarged



Positive (+) **Negative (−)**

White = no contribution of that sign

Fixed head target

$$f(z) = u^{\top} \sum_{j} A(x)_{\mathrm{CLS},j} V(z)_j$$

Path attribution to input coordinate i

$$R_i = (x_i - x_{0i}) \int_0^1 \partial_i f(x_0 + \alpha(x - x_0))\, d\alpha$$

Recorded B0/H10 example chosen for illustration · original cells, no spatial smoothing